

Module 4 : ENERGY SOURCES

Syllabus: - Energy sources:- Basic concepts- Conventional & non-conventional ,solar energy, Fuel cells, Wind energy, Small hydro plants , Bio-fuels, Energy derived from oceans, Geothermal energy .

Energy Sources

Energy is a vital force in our daily life. Energy takes many natural forms: light, heat, sound, electrical energy, kinetic energy, chemical energy, nuclear energy and so on. The need for energy has increased drastically due to population explosion and the modern world needs access to modern energy utilization and conservation mechanisms.

	Renewable: Energy sources that are directly available & can be replenished (made again)
Energy Sources	in a short period of time. Eg:- solar energy, wind energy, geothermal energy,biomass etc.
	Non-Renewable: Energy sources that cannot be replenished in a short period of time.
	Eg:- coal, petroleum , natural gas etc.

Conventional Energy Sources: -These are the commonly used sources of energy and generally include NON-Renewable energy sources.

Non-Conventional Energy Sources: - These are the NOT so common sources of energy and generally include Renewable energy sources.

SOLAR ENERGY

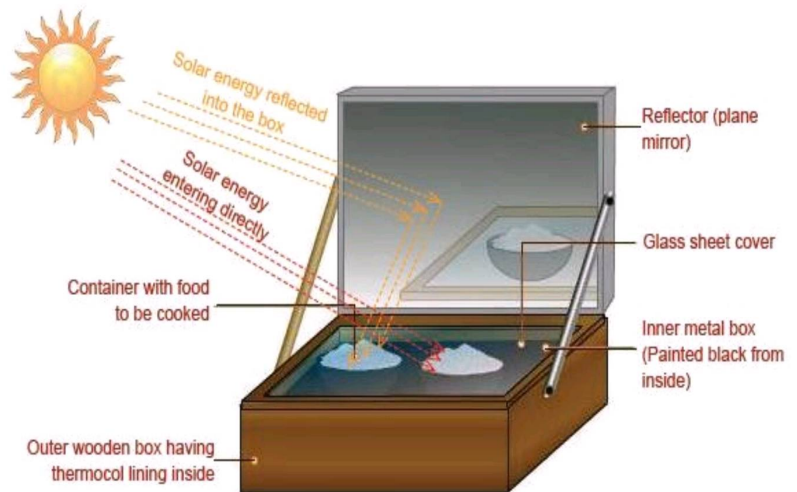
Sun is an enormous source of energy. About 0.1% of total solar energy that reaches the earth can solve all our energy needs. Sun is the basic source all phenomena like ocean waves and currents, wind and photosynthesis. Solar power is a renewable free source of energy that is sustainable and totally inexhaustible. Utilizing solar power for energy requirements needs high amount of initial investment, but the long term benefits make it attractive. Solar energy technology comprises of two different categories: - Thermal conversion & photo conversion.

Applications of solar technologies include:-

- Solar water heating: - Heat from the sun is used to heat water in glass panels on our roof. Water is pumped through pipes in the panel. Using the heat from the sun, water pipes get hot and heat the water without using electricity.
- Solar Cars: It is an electrical vehicle which is recharged from solar energy or sunlight. Solar panels are used on this car that absorb light and then convert it into electrical energy. This electrical energy is stored in batteries used within the car, so that we can drive these vehicles in night time too.

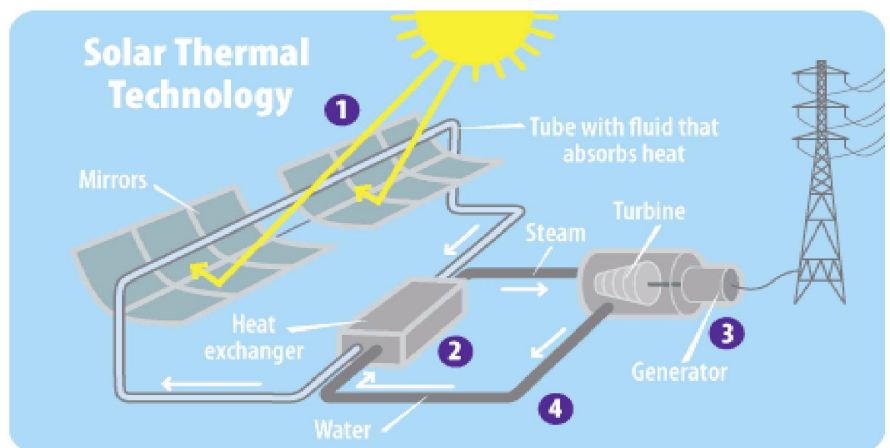
- **Solar Cooker: -**

Solar thermal energy is directly collected and used for cooking. It consists of a well insulated rectangular or square metal box that is blackened on the inner side and fitted with a flat glass cover. When kept in sun, metal box absorbs heat and glass cover reduces its escape, resulting in an increase in temperature inside the box. A well made cooker can achieve an inside temperature of 150° C to 200 °C on sunny day



- **Solar Electricity (Thermal)**

Solar energy is concentrated using collectors and used to heat a fluid which then is used to generate electricity through a conventional heat engine. With the application of solar energy heat of sun can boil water to create steam and rotate turbines.



- **Solar Electricity – (Photovoltaic):** Photovoltaic or PV technology employs solar cells or solar photovoltaic arrays to convert energy from the sun into electricity. Solar cells are made of semiconductor materials. A silicon photovoltaic cell consists of single crystal of p-type silicon with a surface layer of n-type silicon. When light falls on such a p-n junction, electrons and holes move in opposite directions, creating an electric current. Solar cells produce direct current electricity from the sun's rays, which can be used to power equipment or to recharge batteries. Many pocket calculators incorporate a single solar cell, but for larger applications, cells are generally grouped together to form PV modules that are in turn arranged in solar arrays. Solar arrays can be used to power orbiting satellites and other spacecraft, and in remote areas as a source of power for roadside emergency telephones, remote sensing etc.
- Other applications include solar refrigerators and air conditioners, solar desalination, solar furnaces, solar drying etc.

Solar power in India is a fast-growing industry. As of 30 September 2016, the country's solar grid has a cumulative capacity of 8,626 MW (8.63 GW).

Advantages of Solar Energy

- Inexhaustible renewable energy source
- No pollution

---Environmental impacts can be reduced.

Limitations

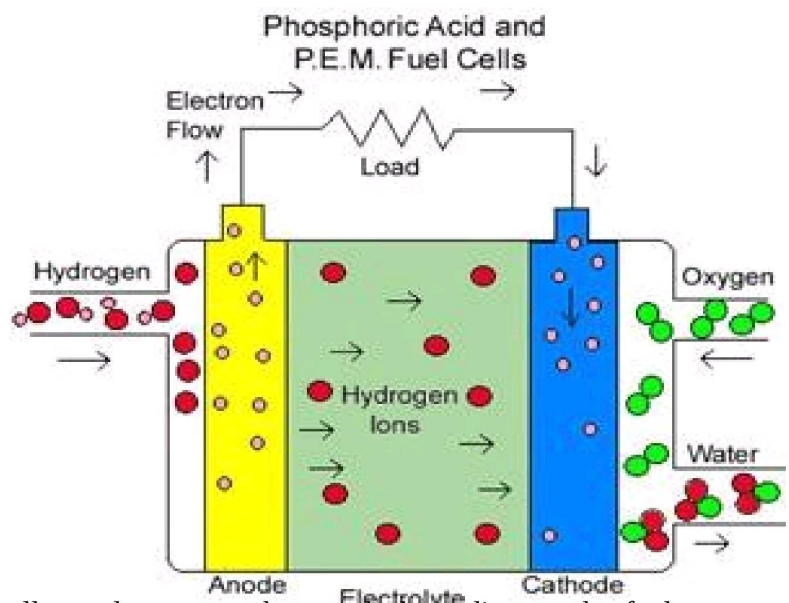
- a) Sunny days are required to capture effective solar energy.
- b) Initial set up cost is high
- c) Batteries should be used to store energy for night and cloudy days, but batteries will be large, heavy and need storage space.

FUEL CELLS

A fuel cell is a device that converts the chemical energy from a fuel into electricity through a chemical reaction of positively charged hydrogen ions with oxygen or another oxidizing agent. Hydrogen is the basic fuel, but fuel cells also require oxygen. There are many types of fuel cells, but they all consist of an anode, a cathode, and an electrolyte that allows positively charged hydrogen ions (or protons) to move between the two sides of the fuel cell. The anode and cathode contain catalysts that cause the fuel to undergo oxidation reactions that generate positively charged hydrogen ions

and electrons. The hydrogen ions are drawn through the electrolyte after the reaction. At the same time, electrons are drawn from the anode to the cathode through an external circuit, producing direct current electricity. At the cathode, hydrogen ions, electrons, and oxygen react to form water. Individual fuel cells produce relatively small electrical potentials, about 0.7 volts, so cells are "stacked", or placed in series, to create sufficient voltage to meet an application's

requirements. In addition to electricity, fuel cells produce water, heat and, depending on the fuel source, very small amounts of nitrogen dioxide and other emissions. The energy efficiency of a fuel cell is 40–60 %.



Applications

- It is used in space crafts.
- Power source in remote areas.
- Can be applicable in both hybrid & electric vehicles

Advantages

- a) It is compact light weight and has no moving parts.
- b) Non Toxic and does not contribute to climatic change
- c) Available and Renewable
- d) Energy conversion efficiency is very high (60%)

Disadvantages

- a) Overall production cost is high.
- b) Hydrogen is very prone to catch fire, or even exploding

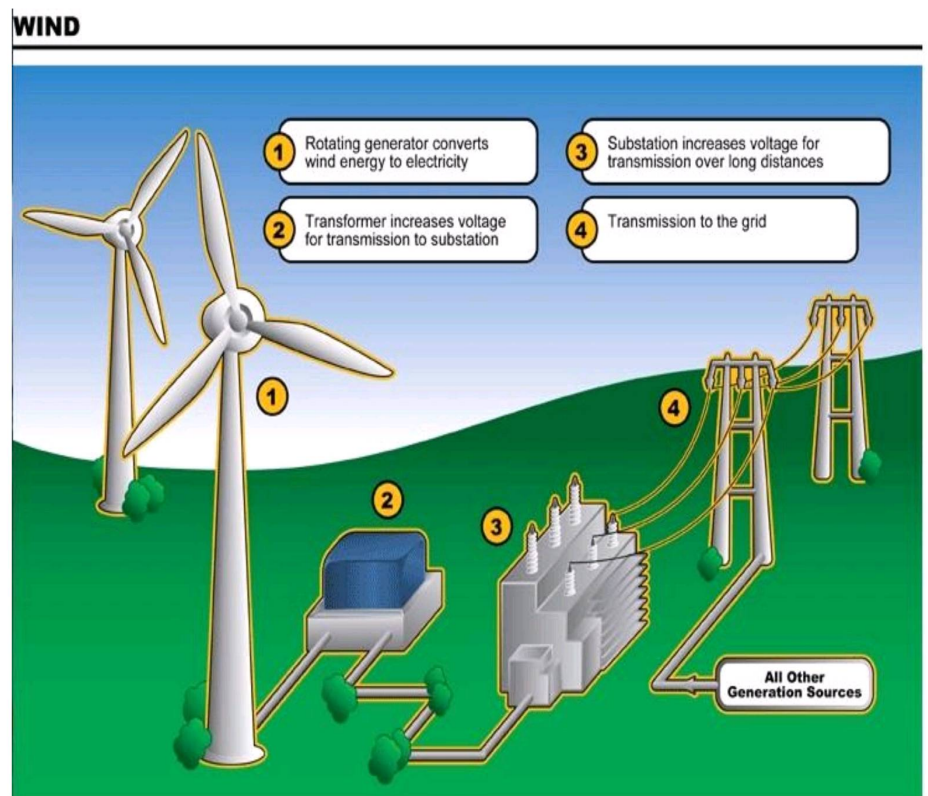
WIND ENERGY

Wind power is the use of air flow through wind turbines to mechanically power generators for electricity. Wind power, as an alternative for burning fossil fuels, is renewable, widely distributed, clean and produce no greenhouse gas emissions during operation. A tall tower with a large propeller on the top forms the basic structure of a wind mill. The wind blows the propeller round which turns a generator to produce electricity. The wind turns the blades, which spin a shaft, which connects to a generator and makes electricity. The kinetic energy of the wind can be used to run wind turbines but the output power depends upon the wind speed. The blades can be angled to cope with varying wind speeds. Also the propeller can turn to face the wind wherever it comes from. The generated electricity can be carried to the bottom of the tower. Through underground cables, it is fed to a transformer and then to a substation. Building several towers together forms a wind farm.

The best place for wind farms are in coastal areas, at the top of rounded hills, open plains and gaps in mountains. Off shore wind mills (wind mills in water bodies) are highly expensive than on shore wind mills. World's leading wind power generators include China, USA, Germany, India etc. In India, wind mills are set up in states of Tamilnadu, coastal regions of Maharashtra, Gujarat, Rajasthan, Kanjikode at Kerala etc .

ADVANTAGES

- clean, pollution less,
- sustainable &
- cost effective fuel source.



Wind turbines can be built on existing farms or ranches. This greatly benefits the economy in rural areas, where most of the best wind sites are found. Farmers and ranchers can continue to work the land because the wind turbines use only a fraction of the land.

Challenges

- Good wind sites are often located in remote locations, far from cities where the electricity is needed.
Transmission lines must be built to bring the electricity from the wind farm to the city.
- Turbines might cause huge noise
- Birds have been killed by flying into spinning turbine blades.
- Wind is not always predictable.

HYDRO POWER

Hydropower or water power is the power derived from the energy of falling water or fast running water, which may be utilized for useful purposes. Hydropower is using water to power machinery or make electricity. Water constantly moves through a vast global cycle, evaporating from lakes and oceans, forming clouds, precipitating as rain or snow, and then flowing back down to the ocean. The energy of this water cycle, which is driven by the sun, can be tapped to produce electricity or for mechanical tasks. Hydropower uses water that is not reduced or used up in the process. Because the water cycle is an endless, constantly recharging system, hydropower is considered a renewable energy.

When flowing water is captured and turned into electricity, it is called hydroelectric power or hydropower. There are several types of hydroelectric facilities; they are all powered by the kinetic energy of flowing water as it moves downstream. Water is allowed to flow into the electricity generating system through a passage called “penstock”. Turbines run due to the force of moving water and generator converts the energy into electricity, which is then fed into the electrical grid to be used in homes, businesses, and by industry.

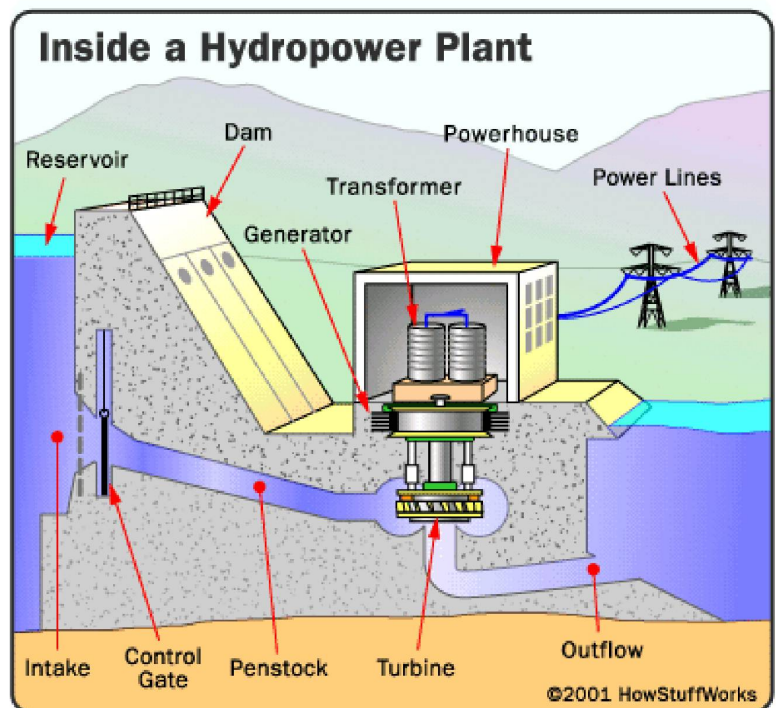
LARGE HYDROPOWER- capacity of more than 30 megawatts (MW).

SMALL HYDROPOWER (SHP) - capacity of 10 MW is accepted worldwide. In India and China, SHP refer to capacity up to 25 MW

MICRO HYDROPOWER – capacity up to 100 KW

ADVANTAGES OF HYDROPOWER

- It's a clean, pollution less, renewable energy source. Relatively short time for construction and operation
- Hydropower efforts produce a number of benefits, such as flood control, irrigation, and water supply.



BIO-FUELS

Bio-fuels are the fuels that are derived from biomass. Biomass is organic matter derived from living, or recently living organisms and includes plant body, wood, food wastes, agricultural crop residue, herbage, animal droppings etc. Biomass has stored chemical energy and it can be converted into useful energy through different processes. This energy conversion results in the formation of different bio-fuels. Energy can be utilized from biomass through direct burning (wood is burnt to produce heat energy), chemical process (biodiesel production), biological process (biogas production), and thermal process.

The energy conversion processes stated above results in the formation of different bio-fuels. Examples of bio-fuels are bio-diesel, bio-gas (Gobar gas), green diesel, bio-ethers, bio-alcohols etc.

- **Bio-diesel**-- It is obtained by the chemical processing of bio-mass. It is produced by chemical processing of vegetable oil and animal fat. The chemical reaction is called as transesterification. Bio-diesels produce less emission than petro diesel, but the vehicle engines should be modified to reduce damage to fuel lines and rubber components.
- **Green diesel**— Produced by refining of vegetable oil. It creates fewer problems than standard diesel and will run in any engine without any modification.
- **Bio-gas** – Mixture of different gases produced by decomposition of organic matter such as garbage, animal waste etc by anaerobic bacteria. This process is called bio-methanation or anaerobic digestion. Bio-gas consists of 65% methane, 30-35% carbon dioxide etc. Methane gas from bio-gas can be captured and burned for cooking and heating. Residue from burning of bio-gas ,called sludge can be dried and used as a fertilizer. Bio-gas can be used to power electricity plants.

First Generation Bio-fuels First generation bio-fuels are made from sugar, starch, or vegetable oil. If the plant or material from which they are generated is used in large quantity, it would have a large impact on the food supply. First generation bio-fuels are the “original” bio-fuels and constitute the majority of bio-fuels currently in use. Eg:- bio-diesel, bio-gas, bio-ethers, bio-alcohols etc

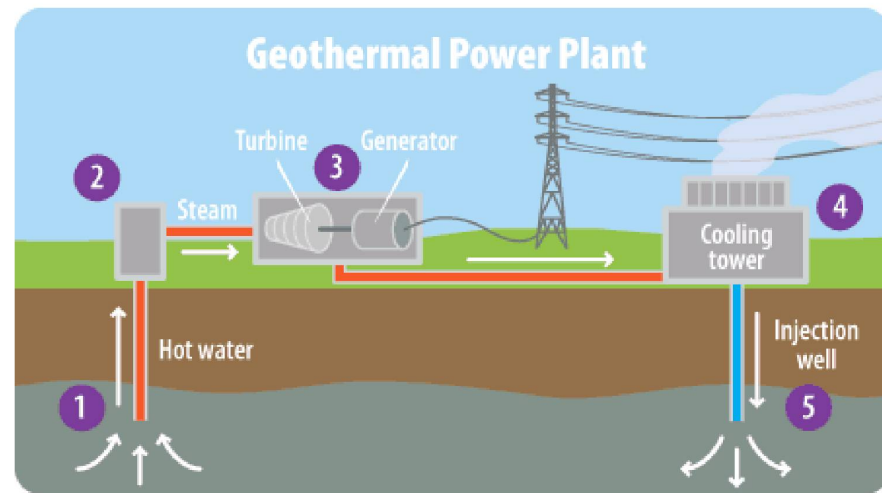
Second Generation Bio-fuels-----Second generation bio-fuels are more sustainable than first generation bio-fuels. In this use, the term sustainable is defined by the large availability of the plant/material, reduced amount of greenhouse gas emissions, less impact on biodiversity and land use (water, food supply, etc.). At this point, most second generation fuels are under development and not widely available for use. Eg:- Bio-hydrogen

Third Generation Bio-fuels -- It has only recently entered the mainstream & refers bio-fuels derived from algae.

Geothermal Energy

Geothermal energy is thermal energy generated and stored in the earth. Geo means earth and thermal means heat. The geothermal energy of the Earth's crust originates from the original formation of the planet and from radioactive decay of materials. Geothermal power is cost-effective, reliable, sustainable, and environmentally friendly. Heat from the earth can be used as an energy source in many ways, from large and complex power stations to small and relatively simple pumping systems.

The Centre of the earth is about 6000°C . Even a few kilometers down, the temperature can be over 250°C . Hot rocks under the earth's surface heat water to produce steam. Holes are drilled down to the hot region and steam comes up. The steam is purified and used to drive turbines and generate electricity. If there is no natural groundwater, we may need to pump



water down to the hot region and convert it into steam. Geothermal energy can be used for:-

- **Geothermal heating:** - hot steam /water under the earth's surface can be brought to the surface and directly used for domestic heating purposes.
- **Geothermal electricity:** - hot steam /water under the earth's surface can be brought to the surface and used to run steam turbines to generate electricity.

Geo thermal energy is an important resource in volcanically active places such as Iceland, New Zealand etc. Two – third of the residential places in Iceland is heated using geothermal energy. Many countries around the world use geothermal energy to produce electric power – USA (3000 MW), Mexico (700 MW), New Zealand (300 MW) etc. In India, studies are going on to find out the possibilities of geothermal plants in Puga Valley (J&K), Tatapani (Chattisgarh), Godavari Basin Manikaran (Himachal Pradesh), Bakreshwar (West Bengal) etc.

Advantages

- a) Renewable, environment friendly
- b) not affected by seasons or day-night cycles
- b) It can be used directly for space heating, hot water generation etc.

Disadvantages

- a) Only available in certain regions
- b) overall efficiency of power production is just 15%

ENERGY FROM OCEANS

70 % of the earth is covered with water and oceans are a store house of energy. Energy from oceans can be utilized through different phenomena: -

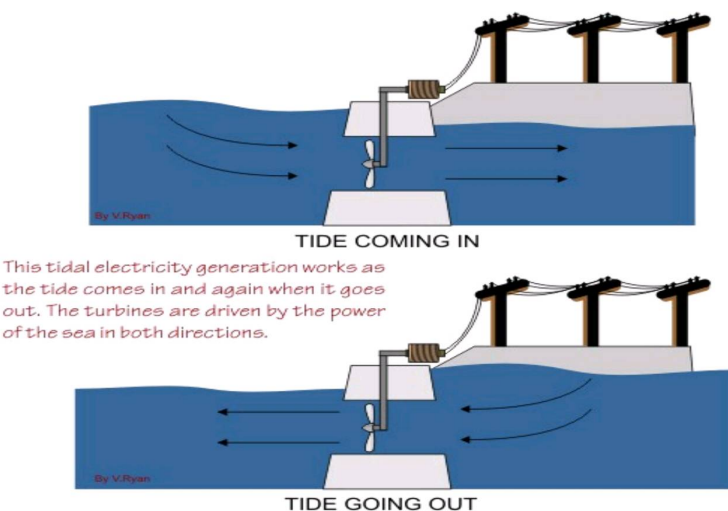
- a) Tidal Energy b) Ocean wave energy c) Ocean Thermal energy conversion (OTEC)

• TIDAL ENERGY

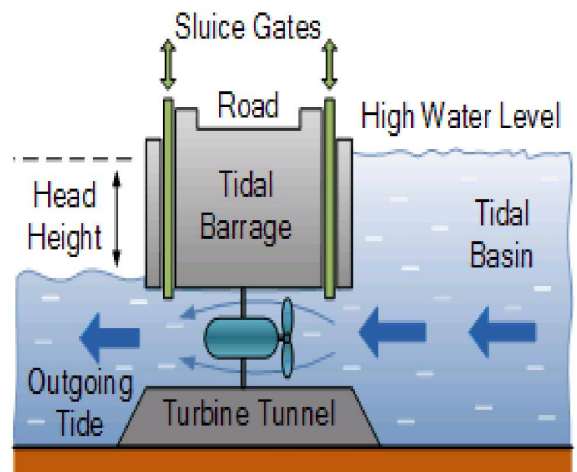
Tides are generated by the relative motion of earth, sun and moon, which interact via gravitational forces. Energy derived from tides is practically inexhaustible and classified as renewable energy source. Here the kinetic energy of moving tidal water is used for energy generation using a turbine. The stronger the tide, the greater is the potential for tidal energy generation.

Categories of tidal power: -

--Tidal stream systems --- kinetic energy of moving water is used to power turbines.



--Tidal Barrage --- Makes use of potential energy difference in height between high tide and low tide.



Tidal power plants exist in Canada, South Korea, France, Russia, China etc . In India, proposals and studies are on the way to construct tidal power plants in Gulf of Kutch, Gulf of Cambay (Gujarat) and Sunder bans (West Bengal) .

• OCEAN WAVE ENERGY

Ocean waves are caused by wind as it blows across the sea. At a wave power station, waves arriving cause the water in the chamber to rise and fall, which means that air is forced in and out of the hole in the top of the chamber. A turbine is placed in this hole, which is turned by the air rushing in and out. The turbine turns a generator producing electricity.

Wave energy

- Can be developed at a greater variety of sites than tidal energy
- The motion of wind-driven waves at the ocean's surface is harnessed and converted from mechanical energy into electricity
- Offshore or coastal

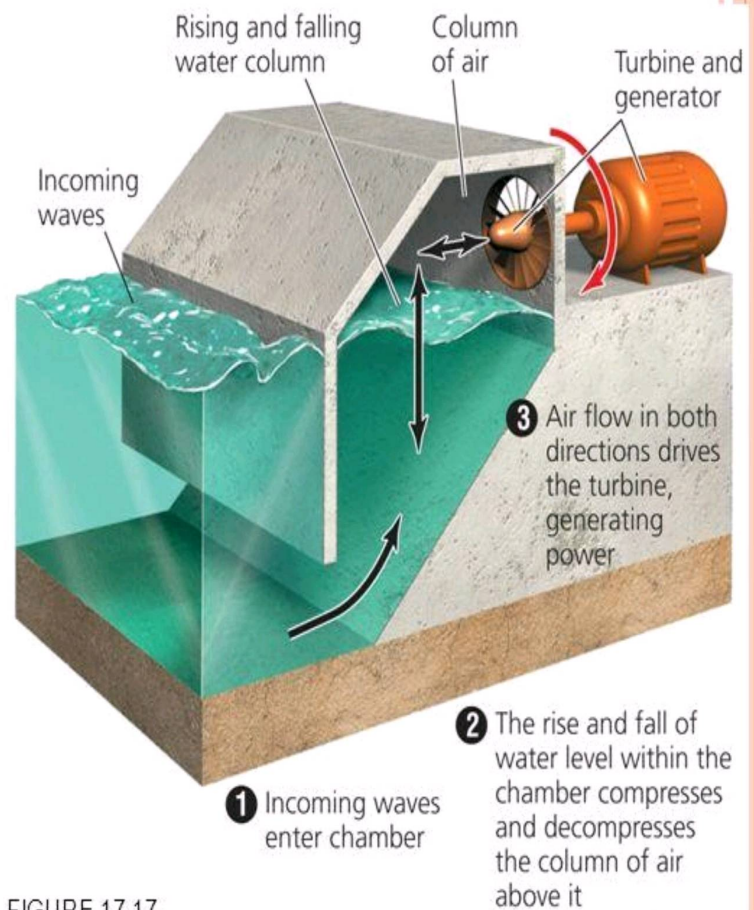


FIGURE 17.17

17-35

In India, Trade wind belts in Arabian Sea and Bay of Bengal can be studied and utilized to look for the possibility of using ocean wave energy.

- **OCEAN THERMAL ENERGY CONVERSION (OTEC)** –Utilizes temperature variation between warmer surface layer and cold bottom layer of oceans to generate energy

Advantages of tidal energy, ocean wave energy

- × Renewable and pollution free.
- × Density of water greater than air so more energy can be produced.
- × Large land area is not required.

Limitations

- × Generation of power depends upon the occurrence of tide, wave etc.
- × There are only few suitable sites for tidal barrages.

